



ABS5321

Issue 6

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Aerospace series

Carbon fibre reinforced epoxy prepreg

Unidirectional tape 180°C curing class

Intermediate modulus fibre

Structural material

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1 Scope

This standard specifies the required characteristics EP-prepreg made from unidirectional intermediate carbon fibre preimpregnated with epoxy or modified epoxy resin for structural applications.

2 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS05-01-000 Part 1	Airbus Material Specification – Carbon fibre reinforced epoxy prepreg/ General requirements – Technical Specification
AIMS05-01-000 Part 2	Airbus Material Specification – Carbon fibre reinforced epoxy prepreg Unidirectional tape / 180°C curing class – Qualification program / Batch release testing – Technical Specification
AIMS05-01-002	Airbus Material Specification – Carbon fibre reinforced epoxy prepreg – Unidirectional tape / 180°C curing class – Intermediate modulus fibre – Structural Material – Material Specification

3 Requirements

3.1 Configuration, dimensions and tolerances

The configuration, dimensions, tolerances and mass shall be in accordance with figure 1, table 1 and table 2. Length, roll inner diameter and mass per roll could be found in the general Technical Specification or in the Individual Product Specification (IPS) if necessary.

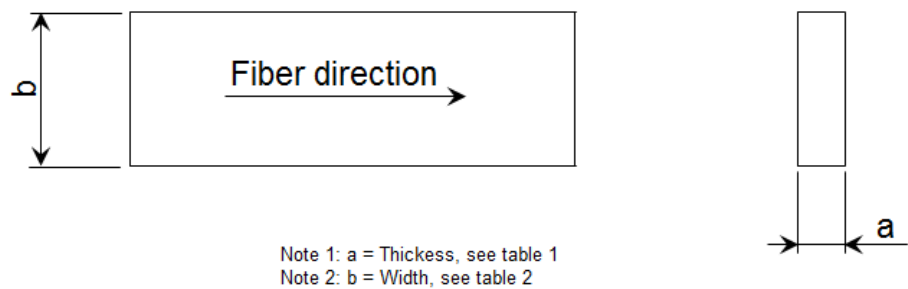


Figure 1: Configuration

Table 1: Dimensions and tolerances

Material ¹⁾	Size Code	Delivery width (b) ²⁾	
		Width (mm)	Tolerance (mm)
All UD prepeg grades (A, B, C, D)	0000 ²⁾	-	-
	0003	3,175	± 0,127
	0006	6,350	
	0025	25	± 0,25
	0150	150	± 0 / - 0,2
	0075	75	± 0 / - 0,75
	0150 ³⁾	150 ³⁾	
	0300	300	
	0600	600	
	1200	1200	
	1219	1219	± 3
Notes: ¹⁾ Length and mass could be found in general MS or IPS if necessary. ²⁾ Size code 0000 means that the width is not defined, any width is permissible. If IPS number is not specified, it is allowed not to write this code. ³⁾ Applicable for AIMS05-01-002A only.			

3.2 Material

The material shall be in accordance with table 2.

Table 2: Materials and mass

Material code ¹⁾	Material	Resin Content (% in weight)	Prepeg areal weight (g/m ²)	Thickness ²⁾ (mm)	
				Nominal (cpt) ³⁾	60% FVC
A	AIMS05-01-002A	34 ± 2	293 ± 10	0,184	0,180
B	AIMS05-01-002B	35 ± 2	412 ± 15	0,260	0,250
C	AIMS05-01-002C	33 ± 1,5	400 ± 12	0,250	0,250
D	AIMS05-01-002D	34 ± 1,5	406 ± 12		
Notes: ¹⁾ For previous designation see table 3. ²⁾ Cured ply thickness may vary depending on specific part geometry and manufacturing process. ³⁾ Nominal cpt is the result of calculation using nominal values for densities, FAW and RC for a zero bleed process.					

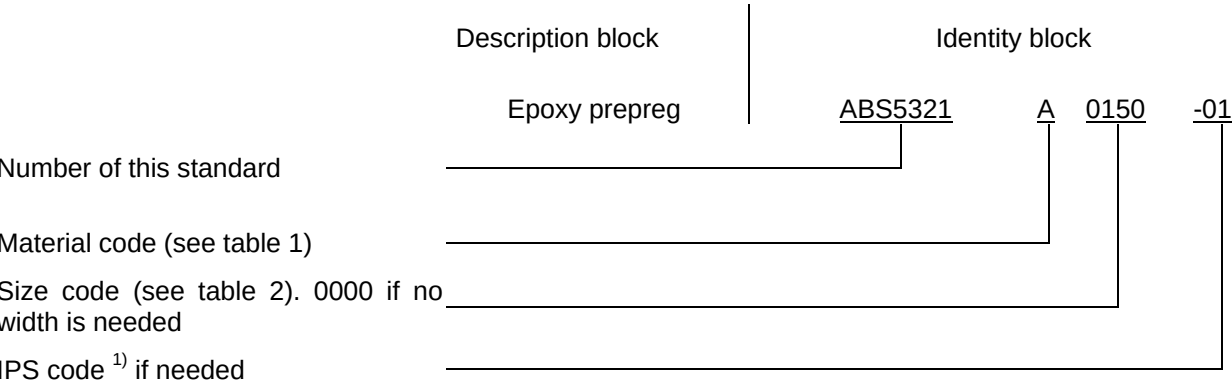
Table 3 : Codes equivalences (Previous codes)

Column 1	Column 2 (previous designation)			
Material identity block	Airbus France Code	Airbus Germany Code	Airbus UK Code	Airbus Spain Code
ABS5321B0000-02	ABS5321A05-02	-	-	-
ABS5321B0000-03	ABS5321A05-03	-	-	-
ABS5321B	-	ABS5321A3	-	-
ABS5321A0000	-	-	-	ABS5321A2X
ABS5321A0150	-	-	-	ABS5321A2A
ABS5321A0300	-	-	-	ABS5321A2B
ABS5321A3175	-	-	-	ABS5321A2C

NOTE: Alternative material identity codes are provided in table 3. The standard material identity code is given in column 1. The corresponding alternatives codes are given in column 2.

4 Designation

This type of standard shall be designated according to the philosophy of the following example:



¹⁾ Blank when no IPS code required. The IPS code is in accordance with the last two digits of the IPS number linked to the AIMS material code.

5 Marking

In accordance with AIMS05-01-000 Part 1

6 Technical specification

In accordance with AIMS05-01-000 Part 1 and Part 2.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 06/00		New standard
2 11/03		Complete revision
3 02/08	Table 1, Table 2	Material code D added, and modification of tolerances for RC and PAW to fit with the average requirements of AIMS05-01-002
4 02/09	Table 2	Size code 0006 (6,35 mm) added
5 01/12	Table 2	Tolerances amended
6 09/13	Table 2	Size code 0075 (75 mm) added